

PLAIN-LANGUAGE GUIDE

SPORTS SCIENCE + WEB TECH

EXECUTIVE OVERVIEW

VMA Calculator: How This Site Works

A clear, non-technical walkthrough explaining how this web application measures aerobic running fitness, performs precision math, and safely organizes athlete records.



Date: March 2026



Tests: Cooper (12m) & Demi-Cooper (6m)



Tech Stack: React 18, Node.js, PostgreSQL

1. What Is This Website and Why Was It Built?

The **VMA Calculator** is a specialized web tool built for running coaches, physical education teachers, competitive athletes, and fitness enthusiasts.

Its mission is simple: **eliminate the math headache and paperwork on the running track**. In the past, a coach timing runners with a stopwatch had to write meters on paper, note who stopped to tie a shoelace, and later spend an evening doing tedious arithmetic to convert meters, seconds, and pause times into usable fitness numbers.

This application automates that entire process in real time. You enter the test numbers, and within a fraction of a second, the site calculates the runner's aerobic speed, assigns a standardized fitness tier, displays every intermediate math step, and lets coaches track improvement across entire seasons.



Instant Calculations

Calculates accurate aerobic speed in milliseconds with real-time live preview as you type.



Long-Term History

Securely tracks individual runners and whole rosters over time with charts and progression trends.



Professional Reports

Generates branded single-athlete PDF certificates and bulk CSV exports for Excel.

2. The Sports Science: What Is VMA and Why Does It Matter?

You don't need a sports physiology degree to understand what this site calculates. It all centers around one metric: **VMA** (*Vitesse Maximale Aérobie*, or **Maximum Aerobic Speed** in English).



The "Car Engine" Analogy

Imagine your body is a hybrid car engine:

- **Aerobic Cruising (Walking / Light Jogging):** Your engine runs on clean oxygen. You can maintain this pace comfortably for hours while chatting.
- **The Maximum Cruising Limit (Your VMA):** As you run faster, your heart and lungs work harder to pump oxygen to your muscles. Eventually, you hit the maximum speed at which your lungs can supply 100% of the oxygen your muscles demand. **That exact running speed is your VMA.**
- **The Redline / Turbo Zone:** If you try to sprint faster than your VMA, your engine runs out of oxygen. Your body switches to emergency anaerobic burning, generating heavy lactic acid, rubbery legs, and total exhaustion within 1 to 2 minutes.



Why Coaches Care: Setting Training Paces Scientifically

Knowing an athlete's VMA (for example, **14.0 km/h**) allows a coach to calibrate every workout:

- **Warm-up & Recovery (60% VMA):** 8.4 km/h
- **Endurance Base (70–75% VMA):** 9.8 – 10.5 km/h
- **Threshold / Tempo (85% VMA):** 11.9 km/h
- **Interval Speedwork (100–110% VMA):** 14.0 – 15.4 km/h

3. The Two Supported Tests

The site supports the two gold-standard field tests used in schools, athletics clubs, and military assessments:

FEATURE	THE COOPER TEST	THE DEMI-COOPER TEST
Duration	12 minutes (720 seconds)	6 minutes (360 seconds)
Origin	Created in 1968 by Dr. Kenneth Cooper for the US Air Force	Shorter, highly practical modern alternative
Goal	Cover maximum possible distance in 12 minutes	Cover maximum possible distance in 6 minutes
Best For	Seasoned runners and endurance athletes	Beginners, school classes, and frequent progress check-ins
Physiology	Tests pacing endurance and aerobic stamina	Runners hit their maximum oxygen uptake (VO ₂ max) right at minute 6
Fatigue & Recovery	High physical toll; requires several days of recovery	Moderate toll; can easily be repeated every month

4. The 5 Fitness Tiers (Color-Coded Classification)

Once VMA is computed in km/h, the application automatically assigns the runner a visual tier:

TIER	VMA RANGE	RATING	WHAT IT MEANS IN PLAIN ENGLISH
Level 1	< 10.0 km/h	Beginner	Starting their running journey; building foundational cardiovascular health.
Level 2	10.0 – 12.0 km/h	Intermediate	Active recreational runner; solid aerobic base for casual 5K/10K runs.
Level 3	12.0 – 14.0 km/h	Good	Regular club runner; strong cardiovascular conditioning and stamina.
Level 4	14.0 – 16.0 km/h	Very Good	Competitive endurance athlete; high aerobic power and sustained speed.
Level 5	> 16.0 km/h	Excellent	Elite / advanced distance runner; exceptional aerobic efficiency.

5. The Math Demystified: The 4-Step Process

Why can't you just divide total meters by total minutes on a phone? Because **in the real world, runners take breathers, pause to tie a shoe, or walk through a stitch.**

How the Site Calculates True Aerobic Velocity:

Step 1: Determine Total Time in Seconds (Cooper = 720 s, Demi-Cooper = 360 s).

Step 2: Calculate Effective Running Time by subtracting pauses:

Effective Time (seconds) = Total Test Time – Stop Time – Walking Time

Step 3: Convert to Standard Units (Hours = seconds ÷ 3,600 | Kilometers = meters ÷ 1,000).

Step 4: Compute VMA in km/h:

VMA (km/h) = Distance in Kilometers ÷ Effective Time in Hours

(Result is rounded cleanly to two decimal places, e.g. 13.09 km/h)

6. Real-World Math in Action: Meet Sarah

To see how these numbers come together, let's walk through a concrete example:

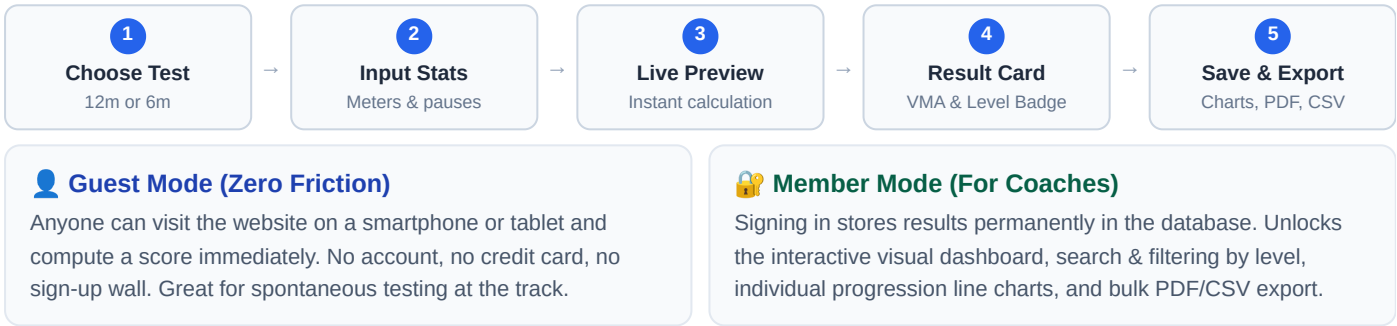
 **Case Study: Sarah's Cooper Test**

Sarah runs the 12-minute Cooper Test. She covers **2,400 meters**, but had to pause for **30 seconds** to tie her shoelace and walked for **30 seconds** when she felt a stitch.

- **Total Test Duration:** 720 seconds (12 minutes)
- **Effective Running Time:** $720 - 30 - 30 = 660$ seconds (11 minutes of pure running)
- **Effective Time in Hours:** $660 \div 3,600 = 0.1833$ hours
- **Distance in Kilometers:** $2,400 \text{ meters} \div 1,000 = 2.4$ km
- **Sarah's True VMA:** $2.4 \div 0.1833 = 13.09$ km/h
- **Assigned Tier:** Level 3 — Good (Solid club runner)

7. The User Journey: How Someone Uses the Site

The website is designed for maximum speed and simplicity on any mobile or desktop screen:



Smart Built-in Features & Reporting Suite

 **Live VMA Preview**
As you type meters and seconds, the site calculates the VMA instantly on screen before you even click submit!

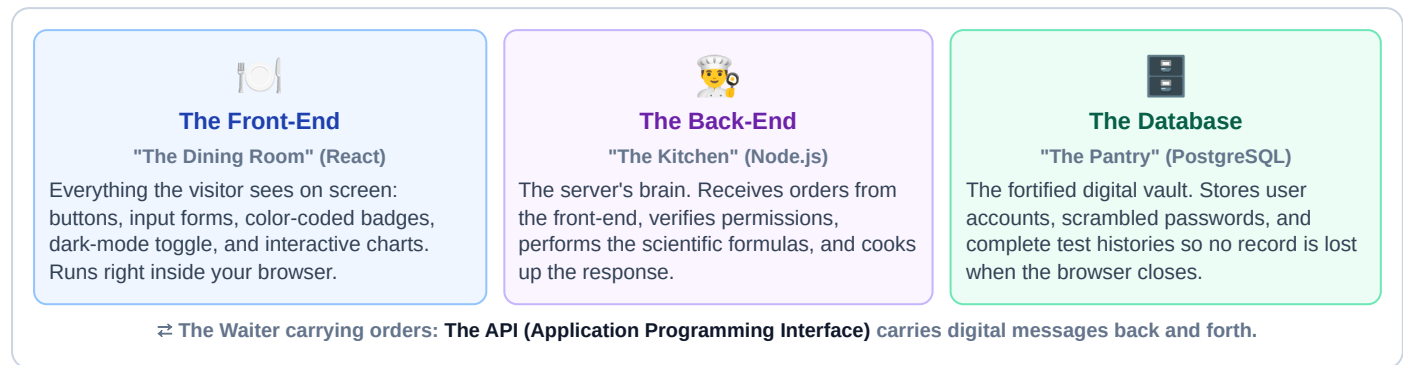
 **Visual Analytics**
Interactive bar and line charts show team level distributions and track individual athlete progress across tests.

 **PDF & CSV Exports**
Generate official printable test certificates for runners or download full spreadsheet rosters for Excel.

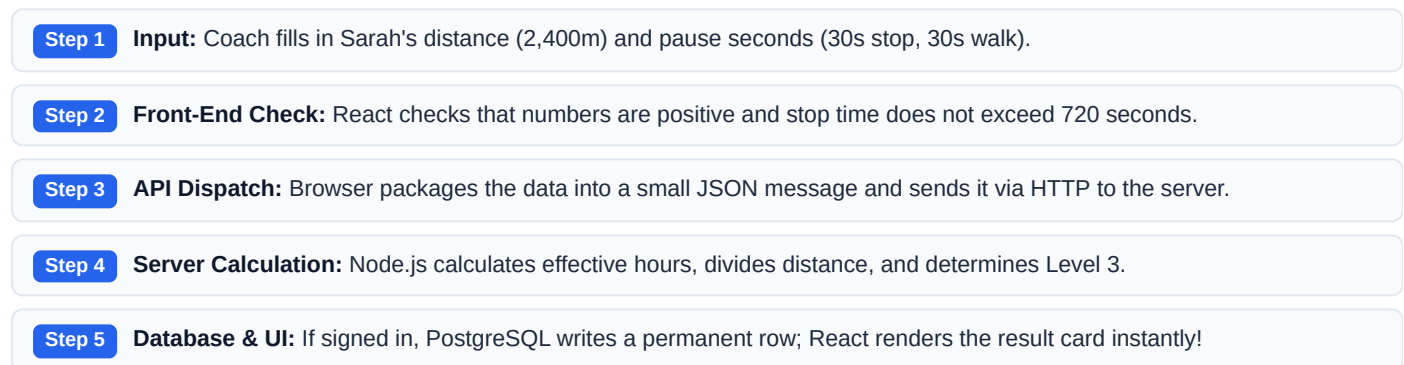
 **Dark Mode Included**
The site features a full dark-mode toggle (sun/moon icon in the header) that saves your preference automatically, perfect for evening track sessions or indoor analysis.

8. Under the Hood: Light Technical Explanation

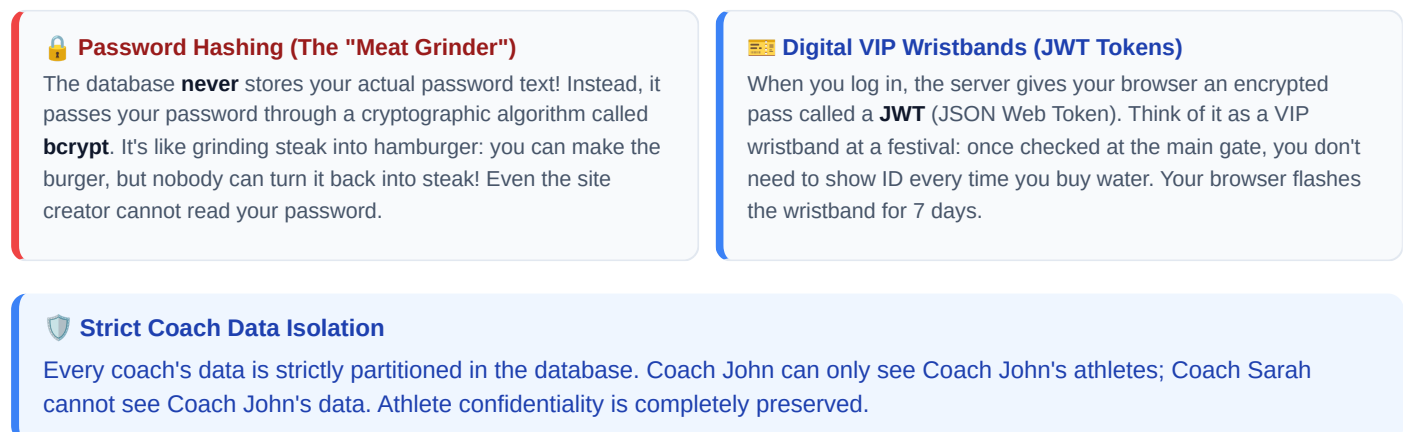
How does the website work from a software perspective? You can think of the entire system as a **busy, high-end restaurant**:



The 5-Step Journey of a Click (What Happens in 50 Milliseconds)



9. Security & Accounts: How Your Data Stays Protected



10. How the Site Reaches Your Phone (Networking Made Simple)

How can an athlete on a sports track open this application on their iPhone or Android when the code is running on a server computer miles away?

1. The Smart Receptionist (Reverse Proxy)

A tiny traffic director program (`proxy.js`) listens on port 8080. When incoming visitors arrive, the receptionist checks what they need:

- "Looking for the web page, buttons, or design?" → Handled by the fast static front-end.
- "Looking to calculate VMA or save a test?" → Forwarded directly to the Node.js API.

2. The Secure Highway (Cloudflare Tunnel)

In the old days, putting a computer on the internet meant punching holes in your office router (port forwarding), which created serious security risks. This project uses **Cloudflare Tunnels**:

- The server establishes a private, encrypted outbound tunnel to Cloudflare's global network.
- Cloudflare gives the app a secure public address (like `https://your-tunnel.trycloudflare.com`).
- Anyone in the world can open that link with full bank-grade HTTPS encryption, while the local computer stays 100% shielded behind its firewall!

11. Quick Reference & Plain-Language Glossary

TERM	WHAT IT MEANS IN ONE SIMPLE SENTENCE
VMA	The maximum running speed where your body can still supply 100% of needed oxygen to muscles.
Effective Time	The true time spent running, calculated by subtracting pauses and walking breaks from test time.
Front-End (React)	The interactive visual screen you tap on your phone or computer browser.
Back-End (Node.js)	The server "kitchen" that validates inputs, applies scientific math, and enforces rules.
Database (PostgreSQL)	The permanent digital filing cabinet storing user accounts, athlete profiles, and test histories.
API	The communication channel carrying questions and answers between the front-end and back-end.
JWT Token	A digital VIP wristband that keeps you securely logged into the site for 7 days.
Password Hashing	Scrambling passwords through a one-way mathematical grinder so they can never be stolen in plain text.
Reverse Proxy	A smart receptionist directing incoming web visitors to either the visual site or the calculation API.
Cloudflare Tunnel	A secure encrypted internet bridge exposing the website worldwide without opening router holes.